

Knowledge Representation and Reasoning Project Report

Semantic Web Application: Basic Linked Data

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Abstract

This report details the development of a Semantic Web application focusing on RDF and Linked Data principles. The project aimed to convert a movies dataset into RDF format, perform meaningful queries using SPARQL, and provide an interactive interface using Streamlit. Despite initial plans to implement advanced Semantic Web concepts, we adapted the scope due to complexity while achieving the core objectives of the project. This document outlines the methodology, challenges, and outcomes of our work.

1. Introduction

The Semantic Web provides a framework to interconnect data across domains using standards such as RDF (Resource Description Framework) and SPARQL. Our project focused on implementing a Semantic Web application in the movies domain to demonstrate the practical application of these technologies. The objective was to process a dataset, model it in RDF, and query it effectively to derive insights while providing an interactive user interface.

2. Project Overview

The project's primary focus was to represent movies, directors, actors, and related meta-data in RDF format. A Streamlit-based application was developed to allow users to interact with the data through search and predefined queries. The key milestones included data preprocessing, RDF generation, and application development.

2.1. Team Members and Roles

The team was divided into three sub-teams based on the proposal:

- **RDF Data Modeling:** Fayaz Noor, Kashif Khan, Aashir Shehzad
- **SPARQL Query Development:** Muhammad Hussain, Sami ur Rehman, Muhammad Huzefa

- **Documentation:** Muhammad Afnan, Ibrahim Aslam

3. Timeline and Workflow

The project was executed over two months, with work divided into weekly intervals. A significant portion of time was dedicated to learning Semantic Web concepts due to the project's complexity.

Week 1–2: Initial Research

- Domain selection and understanding Linked Data and RDF basics.
- Familiarization with SPARQL, Apache Jena Fuseki, and Streamlit.

Week 3–4: Preprocessing and Learning Tools

- Data cleaning and transformation in Python.
- Creation of structured CSV datasets for movies, directors, actors, and countries.

Week 5–6: RDF Data Modeling

- Conversion of the preprocessed CSV data into RDF format using Python scripts.
- Generation of Turtle files to represent data in RDF.

Week 7–8: Application Development

- Development of the Streamlit app for querying and visualizing data.
- Implementation of features like movie search, predefined questions, and recommendations.

4. Methodology

4.1. Data Preprocessing

The movies dataset was cleaned and preprocessed using Python. Redundant and null values were removed, and relevant features were extracted. The preprocessed data was saved as CSV files for use in RDF generation.

4.2. RDF Generation

Using the `rdflib` library in Python, the CSV data was converted into RDF triples. Classes and properties for movies, directors, actors, and countries were defined. The resulting RDF was saved in Turtle format.

4.3. SPARQL Query Development and Apache Jena Fuseki

SPARQL queries were written to extract insights from the RDF data. These queries were executed and tested using `**Apache Jena Fuseki**`, a powerful RDF store and SPARQL endpoint. Fuseki provided a platform for hosting the RDF dataset, enabling interactive querying via a web interface and programmatic access through the Streamlit application.

Queries such as retrieving top movies by budget or finding movies directed by the same director were implemented and optimized in this environment.

4.4. Application Development

A Streamlit-based interface allowed users to:

- Search for movies by title.
- Execute predefined queries.
- View recommendations based on directors and actors.

The application communicated with the hosted RDF data on Fuseki using the SPARQLWrapper Python library, ensuring seamless integration between the backend data and the user interface.

5. Results

The final application successfully showcased the practical implementation of RDF and SPARQL. Users could interactively search and visualize movie data. The project's deliverables included:

- RDF Turtle file representing the dataset.
- Streamlit application for user interaction.
- Comprehensive project documentation.

6. Challenges and Learnings

- **Learning Curve:** Understanding RDF, SPARQL, and tools like **Apache Jena Fuseki** posed an initial challenge. Team members invested significant time learning how to host RDF datasets and query them effectively.
- **Data Complexity:** Representing complex relationships in RDF required careful planning.
- **Technical Issues:** Debugging SPARQL queries in **Fuseki** and ensuring compatibility with Streamlit took significant effort. Integrating SPARQLWrapper to interact with the Fuseki endpoint also required troubleshooting.

7. Conclusion and Future Work

This project provided valuable insights into Semantic Web technologies and their applications. While we deviated from the original proposal due to complexity, we achieved a functional system that demonstrated Linked Data principles. Future work could explore:

- Expanding the dataset with additional domains.
- Enhancing the user interface with more visualization options.
- Implementing advanced SPARQL queries for deeper insights.

8. References

- `rdflib` Python library documentation
- Streamlit official documentation
- TMDB API documentation for fetching movie posters
- Kaggle datasets repository used for preprocessing